AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X

MODEL EXAMINATION PAPER 2027 AND ONWARDS

Computer Science Paper I

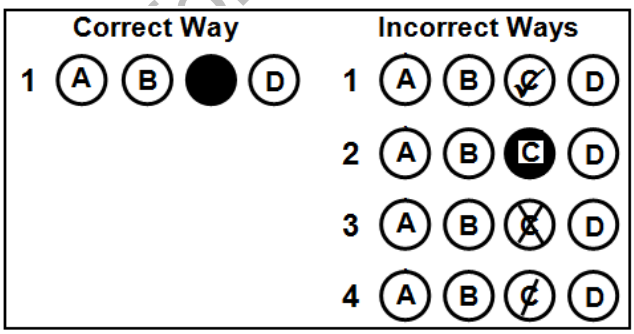
Time: 1 hour 10 minutes Marks: 40

This model paper aims to represent the table of specification with regards to the topics/ content and cognitive demand only.

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INSTRUCTIONS:

1. Check your name and school information on the question paper and Optical Mark Recognition (OMR) sheet. Sign on both if the information is correct.
2. Read each question carefully.
3. Answer all questions on the provided OMR sheet within the allotted time. DO NOT mark your answers on the question paper.
4. There are 100 answer numbers on the OMR sheet. Use answer numbers 1 to 40 only.
5. Each question has four choices: A, B, C, and D. Choose ONE correct answer only. On the OMR sheet, completely fill in the circle corresponding to your choice with a pencil as shown in the example below.



Candidate's Signature

6. If you want to change your answer, ERASE the first answer completely with an eraser, before filling in the new circle. Ensure your marks are dark and clear. Do not make stray marks or leave incomplete fillings as the OMR machine ONLY records what is in the circles.
7. Do not fold, tear, or damage the OMR sheet. Damaged sheets may be scanned incorrectly.
8. You may use a simple calculator if you wish.

1. The MAIN reason for using the hexadecimal number system in computing is to

   A. calculate faster by replacing binary inside processors.
   B. control memory directly by removing byte-order issues.
   C. get better accuracy by fixing rounding in decimal operations.
   D. make reading of binary bits simpler by grouping in sets of four.

2. American Standard Code for Information Interchange (ASCII) is a coding scheme used in computers to

   A. reduce the size of multimedia files during processing.
   B. represent letters, numbers, and symbols in digital form.
   C. store characters from international languages within one system.
   D. protect sensitive information through a secure encryption method.

3. The equivalent decimal (denary) number of the binary number 1011012 is

   A. 18_{10}
   B. 44_{10}
   C. 45_{10}
   D. 63_{10}

4. The 2's complement of the binary number 01101000_2 is

   A. 10010111_2
   B. 10011000_2
   C. 10010110_2
   D. 01101001_2

5. Consider the given algorithm.

```
Step 1: Start
Step 2: A = 18
Step 3: B = 4
Step 4: A = A / B
Step 5: If A > 3 then
          B = B * 2
          Else
          B = B - 1
          EndIf
Step 6: A = A + B
Step 7: Output A
Step 8: Stop
```

   The output of the given algorithm is

   A. 7
   B. 8
   C. 10
   D. 12

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6. Consider the given algorithm.

```
Step 1: Start
Step 2: N = 12
Step 3: If N % 4 == 0 then
        N = N - 1
        Else if N % 3 == 0 then
        N = N + 2
        Else if N < 10 then
        N = N * 2
        Else
        N = N + 1
        EndIf
Step 4: Output N
Step 5: Stop
```

The output of the given algorithm is

A. 11
B. 13
C. 14
D. 24

7. Consider the given algorithm.

```
Step 1: Start
Step 2: S = 0
Step 3: For i = 1 to 4 do
        If i % 2 == 1 then
           S = S + i
        Else
           S = S - 1
        End If
        End For
Step 4: Output S
Step 5: Stop
```

The output of the given algorithm is

A. 0
B. 1
C. 2
D. 3

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8. Consider the given algorithm.

```
Step 1: Start
Step 2: A = 2
Step 3: B = 5
Step 4: Do
        A = A * B
        If A > 20 then
           B = B - 2
        Else
           B = B + 1
        EndIf
        While (A < B * B)
Step 5: Output A - B
Step 6: Stop
```

The output of the given algorithm is

A. 24
B. 34
C. 56
D. 60

9. Consider the given algorithm.

```
Step 1: Start
Step 2: List = [4, 7, 2, 9, 5]
Step 3: Target = 9
Step 4: Found = -1
Step 5: For i = 0 to 4 do
        If List[i] == Target then
           Found = i
           Break
        EndIf
        EndFor
Step 6: Output Found
Step 7: Stop
```

The output of the given algorithm is

A. -1
B. 2
C. 3
D. 4

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10. Consider the given algorithm.

```
Step 1: Start
Step 2: List = [3, 6, 9, 12, 15, 18]
Step 3: Target = 15
Step 4: Low = 1
Step 5: High = 6
Step 6: Found = -1
Step 7: For k = 0 to 3 do
        Mid = (Low + High) / 2
        If List[Mid] == Target then
           Found = Mid
           Break
        Else if List[Mid] < Target then
           Low = Mid + 1
        Else
           High = Mid - 1
        EndIf
        EndFor
Step 8: Output Found
Step 9: Stop
```

The output of the given algorithm is

A. -1
B. 3
C. 4
D. 5

11. Consider the algorithm in which a list is sorted using bubble sort.

```
Step 1: Start
Step 2: List = [5, 1, 4, 2]
Step 3: For i = 0 to 3 do
        If List[i] > List[i + 1] then
           Swap List[i] and List[i + 1]
        EndIf
        EndFor
Step 4: Output List
Step 5: Stop
```

The output of the given list after the first bubble-pass is

A. [1, 4, 2, 5]
B. [1, 5, 4, 2]
C. [5, 4, 2, 1]
D. [1, 2, 4, 5]

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12. The statement that is CORRECT about an if condition is that

A. it block runs only when the condition is true.
B. it always checks two conditions.
C. the else part is compulsory.
D. it is same as nested if.

13. Consider the following code.

```
<button id="btn">Run</button>
<script>
  const msg = "Task completed!";
  document.getElementById("btn").onclick = () => window.alert(msg);
</script>
```

What happens when the given code executes?

A. It will print the text in the console.
B. It will display a pop-up message.
C. It will show the text on the page.
D. It will insert the text in a div.

14. Consider the following code.

```
<script>
    const u = "Hello";
    const v = 42;
    const w = 9007199254740993;
    const x = true;
    let   y;
    const z = null;
    console.log({ u, v, w, x, y, z });
</script>
```

The option that CORRECTLY maps the variables to their data types in order (u, v, w, x, y, z) is

A. u → String, v → Number, w → BigInt, x → Boolean, y → Undefined, z → Null
B. u → String, v → BigInt, w → Number, x → Boolean, y → Null, z → Undefined
C. u → Number, v → String, w → BigInt, x → Boolean, y → Undefined, z → Null
D. u → String, v → Number, w → Number, x → Boolean, y → Undefined, z → String

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15. Consider the given code snippet.

```
<script>
  let total = 10;
  const add = 5;
  total += add;
  total += 2;
  document.write(total);
</script>
```

The output of the given code snippet is

A. 10
B. 12
C. 15
D. 17

16. Consider the given code snippet.

```
<script>
  let score = 50;
  score -= 10;
  score += 5;
  score /= 5;
  score *= 4;
  document.write(score);
</script>
```

The output of the given code snippet is

A. 36
B. 40
C. 45
D. 50

17. Consider the given code snippet.

```
<script>
  const a = 7, b = 10;
  const r1 = a <= b;
  const r2 = a >= b;
  document.write(r1 + " " + r2);
</script>
```

The output of the given code snippet is

A. true true.
B. true false.
C. false true.
D. false false.

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18. Consider the given code snippet.

```
<script>
  const loggedIn = false, hasToken = true;
  const canAccess = loggedIn || hasToken;
  document.write(canAccess);
</script>
```

The output of the given code snippet is

A. null.
B. true.
C. false.
D. undefined.

19. Consider the given code snippet.

```
<script>
  const values = [2, 4, 1, 6, 3];
  let printed = 0;
  for (let i = 0; i < values.length; i++)
   {
    if (values[i] >= 4)
    {
      if (printed > 0) document.write(" ");
      document.write(values[i] * 2);
      printed++;
    }
  }
</script>
```

The output of the given code snippet is

A. 4 6
B. 8 12
C. 2 4 6
D. 8 12 6

20. A global messaging platform integrates Artificial Intelligence (AI) into its services. Users from different countries communicate seamlessly as the system instantly displays messages into their native languages, while also filtering unwanted content.

Based on the given scenario, the AI enriching communication by

A. facilitating national collaboration.
B. supporting secure message delivery.
C. enhancing real-time message conversion.
D. improving one's personal grooming skills.

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21. A retail company applies data mining to its customer purchase records. The system uncovers that customers who buy baby products often purchase household cleaning supplies within the same week.

With reference to the given scenario, the data mining technique applied to uncover purchasing relationships through

A. identifying sequential selling behaviour across product categories.
B. developing predictive analytics for future shopping trends.
C. enhancing clustering models for customer segmentation.
D. discovering association rules in transactional datasets.

22. A national retailer collects billions of transaction records each day. An analyst proactively identifies purchasing trends across different regions and makes necessary adjustments to inventory.

Based on the given scenario, the importance of big data on managing large datasets is to

A. enhance storage capacity.
B. uncover meaningful patterns.
C. support collaboration among teams.
D. improve visualisation of information.

23. A bank includes a machine learning model in its transaction monitoring system. The model identifies unusual spending patterns, such as sudden high amount purchases at unfamiliar locations, and sends alerts to the concerned person for further investigation.

The machine learning skill displayed in the given scenario is

A. automation in financial services.
B. unsupervised learning through clustering.
C. AI enabled systems learning from large data.
D. fraud detection through inconsistency identification.

24. During group work, learners use a shared online document while respecting each other's contributions, avoiding inappropriate comments, and ensuring everyone has access to it.

Based on the given scenario, the learners show the act of responsible digital citizenship by

A. promoting respectful collaboration in digital places.
B. enhancing productivity through real time editing features.
C. increasing communication with digital collaboration tools.
D. supporting self-improvement with interactive online applications.

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25. In a school, a student downloaded copyrighted software from a free file-sharing website and shared it with friends. The school later warns the student about consequences.

With reference to the given situation, it can be inferred that the student has an ethical issue of

A. violating intellectual property laws by illegally copying or using software.
B. prioritising technical efficiency by paying attention to legal requirements.
C. ignoring the value of using a combination of different learning tools.
D. continuing to use outdated approaches for sharing software.

26. A university student spends more than 15 hours every day on his/ her laptop for classes and entertainment, and has begun dealing with frequent headaches and trouble sleeping.

Based on the given situation, the university student is suffering from a health risk that is

A. lower social activity.
B. loss of discomfort.
C. occasional stress.
D. physical harm.

27. The statement that represents cyberbullying in an online space by

A. using digital platforms to intimidate others intentionally.
B. arguing different viewpoints in an online conversation strongly.
C. sharing someone's original work online without their permission.
D. restricting access to certain online sites due to technical limitations.

28. In digital technology, the access divide refers to

A. variation in income levels affecting the affordability of ICT tools.
B. difference in cultural approaches toward technology use.
C. unequal availability of devices and internet connectivity.
D. unequal distribution of digital skills among people.

29. In a rural district, schools struggle because many households lack reliable internet and students cannot join online classes. Teachers report that even when devices are donated, poor connectivity prevents meaningful participation.

The EFFECTIVE strategy that can be applied by the government to reduce the digital divide in the given situation is by

A. offering workshops that prepare users, yet failing to improve internet access.
B. organising awareness campaigns about the importance of online learning.
C. expanding broadband infrastructure with affordable internet packages.
D. encouraging students to share devices among peers during lessons.

30. Digital literacy is referred as the

A. infrastructure that enables internet connectivity.
B. combination of computer and network system.
C. ability to use and evaluate digital information.
D. financial capacity to afford digital tools.

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31. A multinational team needs to coordinate daily tasks, share files instantly, and avoid miscommunication while working remotely.

The effective digital tool to improve communication in the given situation is the use of

A. documentation tool with antivirus software for protection against malware.
B. tracking tool to track tasks with printed reports shared on daily basis.
C. email chains for all updates including reports, regardless of urgency.
D. video conferencing platforms with integrated chat and file sharing.

32. A frequent online shopper worries about identity theft and unauthorised access to accounts.

Based on the given situation, the effective digital technology to enhance security is by

A. associating email alerts with accounts in case of compromising.
B. enabling two-factor authentication for all accounts.
C. keeping separate account for online purchasing.
D. relying solely on browser autofill for logins.

33. A university student struggles to cope multiple assignments, deadlines, and group projects.

The digital technology to solve the given problem effectively by using

A. memory recalling apps to alert about assignments and project frequently.
B. social apps for discussion of assignments and projects to keep updated.
C. task management apps with reminder and collaboration features.
D. notes app to write all task and keep it as a personal diary.

34. A marketing team struggles with version control when multiple members edit campaign documents simultaneously.

The technology that would effectively optimise productivity in the given scenario is by

A. emailing updated files back and forth among team for secure communication.
B. using cloud-based collaboration platforms with live editing rights.
C. saving files on individual desktops with synchronisation.
D. recording and sharing drafts for review and corrections.

35. A professional salesperson regularly uses public Wi-Fi at airports and cafes to check email and access online banking. He/ she is concerned that hackers will intercept his/ her sensitive data.

Based on the given context, the efficient way to protect information is to

A. activate security features such as biometrics login.
B. install a password manager to generate strong credentials.
C. use a secure mobile payment app with built-in encryption.
D. use a Virtual Private Network (VPN) to encrypt all networks.

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36. A university launches a mobile app to improve connectivity with students by giving real time updates on class schedules, buses, and important notices. The observed data shows that the students mostly use this app only for the bus tracking feature.

Based on the given situation, it can be concluded that the mobile app is

A. more effective due to limited use of all the features.
B. completely effective since one feature is used often.
C. ineffective because students are not using some of its features.
D. partially effective because it works for transport features only.

37. A business plan that integrates insights generated through design thinking shows that the

A. assumptions can lead to decisions.
B. user ideas are aligned with the strategy.
C. financial data is obtained with research.
D. innovation is separated from the planning.

38. Financial knowledge, communication, and good decision making skills enable an entrepreneur to

A. focus only on product ideas.
B. manage daily operations smoothly.
C. avoid learning from customers' feedback.
D. dependent only on research for business growth.

39. While creating a business plan, an entrepreneur outlines products or services that help him to

A. focus on product pricing only.
B. skip the features of the product.
C. define clearly what value is being offered.
D. avoid providing services to the customers.

40. A small craft brand runs social media pages but gets very low engagement.

Based on the given information, the strategy that can improve audience engagement by

A. creating interactive reels.
B. uploading content inconsistently.
C. running paid ads without analysing metrics.
D. posting product photos without offering any context.

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END OF PAPER

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